

Azure Migrate for Terraform – User Guide

⚠ Preview Version – This module is currently in preview. Features and APIs may change before the stable release.

Overview

The Azure Migrate for Terraform module provides infrastructure as code (IaC) capabilities for Azure Migrate, enabling automated discovery, replication, and migration of virtual machines to Azure Local. This module supports:

- VMware to Azure Local migrations
- Hyper-V to Azure Local migrations

Key Features

- Discovery: Automatically discover VMs from VMware or Hyper-V environments
- Infrastructure Setup: Initialize replication vaults, policies, and extensions
- Replication: Enable VM replication with granular disk and NIC configuration
- Monitoring: Track replication jobs and protected item status
- Migration: Perform planned failovers (migrations) to Azure Local
- Cleanup: Remove replication when no longer needed

Prerequisites

Required Software

Component	Version
Terraform	>= 1.9
AzAPI Provider	~> 2.4

Azure Requirements

- Azure Subscription with appropriate permissions
- Resource Group where resources will be deployed
- Azure Migrate Project (can be created by this module)
- Azure Local Cluster with: custom location configured, logical networks set up, storage paths defined
- Source Environment: VMware vCenter or Hyper-V host with an Azure Migrate appliance deployed and registered

Required Permissions

The executing identity needs the following permissions:

- Contributor on the resource group
- User Access Administrator (for role assignments)

Quick Start

1. Configure the Provider

```
terraform {
  required_version = ">= 1.9"

  required_providers {
    azapi = {
      source = "Azure/azapi"
      version = "~> 2.4"
    }
  }
}

provider "azapi" {}
```

2. Add the Module from the Registry

```
module "migrate" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerm"
  version = "<VERSION>" # Pin to the latest compatible version

  # ... configuration ...
}
```

Find the latest version at: <https://registry.terraform.io/modules/Azure/avm-ptn-azure-local-migrate/azurerm/latest>

3. Authenticate

```
az login
az account set --subscription "<your-subscription-id>"
```

4. Use the Module

Each step in the migration requires a separate module invocation with a different `operation_mode`. See the detailed sections below.

Operation Modes

The module's behavior is controlled by the `operation_mode` variable. Valid values are:

Mode	Description
<code>create-project</code>	Create a new Azure Migrate project with all solutions
<code>discover</code>	List discovered VMs from the source environment
<code>initialize</code>	Set up replication vault, policy, extension, and storage
<code>replicate</code>	Start replication for a specific VM
<code>jobs</code>	Query replication job status
<code>list</code>	List all protected (replicating) items in a vault
<code>get</code>	Get detailed status of a specific protected item
<code>migrate</code>	Perform planned failover (final migration)
<code>remove</code>	Disable and remove replication for a protected item

Note: The source environment is selected with `source_machine_type` ("VMware" or "HyperV"). The module derives the replication instanceType (VMwareToAzStackHCI / HyperVToAzStackHCI) internally — there is no `instance_type` input.

1. Create Project

Creates a new Azure Migrate project with the required solutions (Assessment, Discovery, Migration, DataReplication) and assigns the necessary role to the project's managed identity.

```
module "create_project" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerms"
  version = "0.1.2"

  location          = "my-location"
  name              = "create-project"
  parent_id         = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
  create_migrate_project = true
  operation_mode     = "create-project"
  project_name      = "my-migrate-project"
  connectivity_method = "Public-endpoint" # or "Private-endpoint"
  tags = {
    Environment = "Production"
  }
}
```

Key inputs:

- `create_migrate_project` — Must be true
- `project_name` — The name for the new Azure Migrate project
- `location` — Required in this mode (there is no existing project to read it from)
- `connectivity_method` — "Public-endpoint" (default) or "Private-endpoint"

Key outputs:

Commented [TB1]: What about resource group? And it won't create it automatically either, you have to provide already created one?

Commented [SA2R1]: That's the `parent_id` input, that's the naming convention we have to follow. Perhaps we add a clarifying section in the doc?

Commented [TB3]: What would this be, like "eastus"?

Commented [TB4R3]: We should list the supported ones

Commented [TB5R3]: | -----
| {
| "error": {
| "code": "LocationNotAvailableForResourceType",
| "message": "The provided location 'eastus' is not available for resource type 'Microsoft.Migrate/migrateprojects'. List of available regions for the resource type is 'centralus,eastasia,southeastasia,northeurope,westeurope,westus2,australiasoutheast,uksouth,ukwest,canadacentral,centralindia,southindia,japaneast,japanwest,brazilsouth,koreasouth,koreacentral,francecentral,switzerlandnorth,australiaeast,uaenorth,southafricanorth,germanywestcentral,norwayeast,jioindiawest,swedencentral,qatarcentral,polandcentral,italynorth,israelcentral,spaincentral,mexicocentral,newzealandnorth,indonesiacentral,malaysiawest,chilecentral,australiaeast,belgiumcentral,denmarkeast,centraluseuap,eastus2euap'.
| }
| }

can we "Azure Local migrations" use both?

Commented [SA6R3]: We have to choose from the accepted regions

- `migrate_project_id` — The ARM resource ID of the created project

Note: Skip this step if you already have an Azure Migrate project. The module works with existing projects, and in all other modes location is read from the project when omitted.

2. Discover Servers

Queries Azure Migrate to list all discovered VMs in your source environment. Read-only; retrieves machine name, IP addresses, OS, boot type, and disk IDs.

```
module "discover" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerm"
  version = "0.1.2"

  name           = "discover"
  parent_id      = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
  operation_mode = "discover"
  project_name   = "my-migrate-project"
}
```

Optional inputs:

- `appliance_name` — Filter by a specific appliance (site). If omitted, all VMs in the project are returned.
- `source_machine_type` — "VMware" (default) or "HyperV"

Key outputs:

- `discovered_servers` — Filtered list (index, machine name, IPs, OS, boot type, OS disk ID)
- `discovered_servers_count` — Number of discovered servers with discovery data
- `total_machines_count` — Total machines including those without discovery data
- `discovered_servers_raw` — Full API response for debugging

Example output:

```
discovered_servers = [
  {
    index           = 1
    machine_name    = "web-server-01"
    ip_addresses    = ["10.0.1.10", "10.0.1.11"]
    operating_system = "Windows Server 2019"
    boot_type       = "EFI"
    os_disk_id      = "6000C290-a4d0-e5ea-bad5-4e993df22e3b"
  },
  # ...
]
```

Tip: Use the `os_disk_id` from discovery output as input for the replicate step.

3. Initialize Replication Infrastructure

Sets up all infrastructure required before replication can begin. Safe to re-run: if replication was already configured it validates that the project is in a good state (vault, extension, fabrics already exist) without overwriting.

```
module "initialize" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerem"
  version = "0.1.2"

  name           = "initialize"
  parent_id      = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
  operation_mode = "initialize"
  project_name   = "my-migrate-project"
  source_machine_type = "VMware"

  # Appliance names – fabrics are auto-discovered from these
  source_appliance_name = "src-appliance"
  target_appliance_name = "tgt-appliance"
}
```

Fabric auto-discovery: Provide `source_appliance_name` and `target_appliance_name` and the module discovers the corresponding replication fabrics automatically. There are no `source_fabric_id` / `target_fabric_id` inputs; the resolved IDs are exposed as outputs.

Key outputs:

- `replication_vault_id` — Vault ARM ID
- `replication_policy_id` — Policy ARM ID
- `replication_extension_id` / `replication_extension_name` — Extension ARM ID / name
- `cache_storage_account_id` / `cache_storage_account_name` — Cache storage account
- `source_fabric_id` / `target_fabric_id` — Resolved fabric IDs
- `source_fabric_discovered` / `target_fabric_discovered` — Discovered fabric details
- `replication_fabrics_available` — All fabrics in the resource group (troubleshooting)
- `replication_vault_identity` — Managed identity principal ID of the vault

Brownfield: If replication infrastructure already exists, the module detects it and skips creation — required settings are validated instead of recreated.

4. Replicate VMs

Starts replication of a source VM to Azure Local by creating a "protected item" in the replication vault.

Default User Mode (single OS disk / single NIC)

```
module "replicate" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerem"
```

```

version = "0.1.2"

name           = "replicate-vm"
parent_id      = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
operation_mode = "replicate"
project_name   = "my-migrate-project"
source_machine_type = "VMware"

# VM to replicate
machine_id = "/subscriptions/.../machines/<machine-guid>" # Use the sdsArmId
parameter from the discover module output

# OS disk (from discover output) – size is derived automatically
os_disk_id = "6000C290-a4d0-e5ea-bad5-4e993df22e3b"

# Network (simple mode – single target logical network)
target_virtual_switch_id = "/subscriptions/.../logicalnetworks/my-lnet"

# Target VM configuration
target_vm_name      = "my-vm-migrated"
target_resource_group_id = "/subscriptions/.../resourceGroups/target-rg"
target_storage_path_id = "/subscriptions/.../storagecontainers/my-storage"
target_hci_cluster_id = "/subscriptions/.../clusters/my-cluster"
custom_location_id   = "/subscriptions/.../customLocations/my-cl"

# Appliance info – replication vault, extension, policy, fabrics,
source_appliance_name = "src-appliance"
target_appliance_name = "tgt-appliance"
}

```

Commented [TB7]: USE "sdsArmId".

Power User Mode (multiple disks / NICs)

For VMs with multiple disks or NICs, use `disks_to_include` and `nics_to_include`:

```

module "replicate" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerem"
  version = "0.1.2"

  # ... (common configuration as above, minus os_disk_id / target_virtual_switch_id)
  ...

  disks_to_include = [
    {
      disk_id      = "6000C290-a4d0-e5ea-bad5-4e993df22e3b"
      disk_size_gb = 40
      disk_file_format = "VHDX"
      is_os_disk    = true
      is_dynamic     = true
    },
    {

```

```

        disk_id           = "6000C29a-bcb7-a62b-7ed0-0c78f3dc1f80"
        disk_size_gb      = 100
        disk_file_format  = "VHDX"
        is_os_disk        = false
        is_dynamic         = true
    }
}

nics_to_include = [
{
    nic_id           = "4000"
    target_network_id = "/subscriptions/.../logicalnetworks/prod-lnet"
    selection_type    = "SelectedByUser"
}
]
}

```

Key outputs:

- `protected_item_id` — ARM ID of the created protected item (needed for `get/migrate/remove`)
- `protected_item_name` — Name of the protected item
- `replication_state` — Current replication health

5. Monitor Replication Jobs

Queries replication job status. List all jobs or retrieve a specific job by name.

```

# List all jobs
module "jobs" {
    source = "Azure/avm-ptn-azure-local-migrate/azurerem"
    version = "0.1.2"

    name           = "replication-jobs"
    parent_id      = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
    operation_mode = "jobs"
    project_name   = "my-migrate-project"
}

# Get a specific job
module "specific_job" {
    source = "Azure/avm-ptn-azure-local-migrate/azurerem"
    version = "0.1.2"

    # ... same as above, plus:
    job_name = "my-specific-job-name"
}

```

Key outputs:

- replication_jobs — Summary of all jobs (name, state, VM, timestamps)
- replication_jobs_count — Total number of jobs
- replication_job — Detailed info for a specific job (tasks and errors)
- vault_id_for_jobs — Vault ID used for the query

6. List Protected Items

Lists all protected (replicating) items in a replication vault.

```
module "list" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerm"
  version = "0.1.2"

  name          = "list-items"
  parent_id     = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
  operation_mode = "list"
  project_name  = "my-migrate-project" # vault auto-discovered
}
```

Key outputs:

- protected_items_summary — Summary (name, ID, state, health, source/target VM names)
- protected_items_count — Total number of protected items
- protected_items_list — Complete raw list
- protected_items_by_state — Items grouped by protection state
- protected_items_by_health — Items grouped by replication health
- protected_items_with_errors — Items that have health errors

7. Get Protected Item Details

Retrieves detailed status for a specific protected item, by ID or by name.

```
# By ID
module "get_by_id" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerm"
  version = "0.1.2"

  name          = "get-item"
  parent_id     = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
  operation_mode = "get"
  project_name  = "my-migrate-project"
  protected_item_id = "/subscriptions/.../protectedItems/<item-name>"
}

# By name
module "get_by_name" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerm"
  version = "0.1.2"
```

```
# ... same as above, but use protected_item_name instead:
protected_item_name = "my-protected-item"
}
```

Key outputs:

- `protected_item` — Complete protected item details
- `protected_item_summary` — Key fields: state, health, allowed jobs, source/target names
- `protected_item_custom_properties` — Fabric-specific details, disk config, network settings
- `protected_item_health_errors` — Any health errors

8. Migrate (Planned Failover)

Performs the final migration by executing a planned failover, creating the VM on the target Azure Local cluster.

```
module "migrate" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerm"
  version = "0.1.2"

  name          = "migrate-vm"
  parent_id     = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
  operation_mode = "migrate"
  target_object_id = "/subscriptions/.../protectedItems/<item-name>"
  shutdown_source_vm = true # Recommended for production
}
```

Key inputs:

- `target_object_id` — The protected item ARM ID (from replicate or list outputs)
- `shutdown_source_vm` — true to shut down the source VM before failover (recommended for data consistency)

Key outputs:

- `migration_status` — Operation status, source/target VM names
- `migration_protected_item_details` — Details of the item being migrated
- `migration_validation_warnings` — Health warnings detected before migration
- `migration_operation_details` — Raw async operation response (for job tracking)

Important: The protected item must be in a Protected state with PlannedFailover in its `allowedJobs` before migration can proceed. Use the `get` operation to verify readiness.

9. Remove Replication

Disables and removes replication for a protected item.

```
module "remove" {
  source = "Azure/avm-ptn-azure-local-migrate/azurerem"
  version = "0.1.2"

  name          = "remove-replication"
  parent_id     = "/subscriptions/<sub-id>/resourceGroups/<rg-name>"
  operation_mode = "remove"
  target_object_id = "/subscriptions/.../protectedItems/<item-name>"
  force_remove  = false # Set true only if normal removal fails
}
```

Key inputs:

- `target_object_id` — The protected item ARM ID to remove
- `force_remove` — Force removal even if the item is in an inconsistent state

Key outputs:

- `removal_status` — Operation status and confirmation message
- `protected_item_details` — Details of the item before removal
- `removal_operation_headers` — Async operation headers for job tracking

Caution: Setting `force_remove = true` may leave resources in an inconsistent state. Use it only as a last resort.

Configuration Modes

Default User Mode

For VMs with a single OS disk and a single target network:

Variable	Description
<code>os_disk_id</code>	OS disk identifier from discovery (size auto-derived)
<code>target_virtual_switch_id</code>	Target logical network ARM ID

Power User Mode

For VMs with multiple disks and/or NICs:

Variable	Description
<code>disks_to_include</code>	List of disk objects (<code>disk_id</code> , <code>disk_size_gb</code> , <code>disk_file_format</code> , <code>is_os_disk</code> , <code>is_dynamic</code>)
<code>nics_to_include</code>	List of NIC objects (<code>nic_id</code> , <code>target_network_id</code> , <code>selection_type</code>)

Power user mode takes priority — when disks_to_include is non-empty, os_disk_id is ignored; likewise nics_to_include overrides target_virtual_switch_id.

Supported Migration Paths

Source	Target	source_machine_type	Derived instanceType
VMware vCenter	Azure Local	VMware	VMwareToAzStackHCI
Hyper-V	Azure Local	HyperV	HyperVToAzStackHCI

Set source_machine_type to match your source environment. This controls fabric discovery, replication extension configuration, and the failover API used.

Variable Reference

Required Variables

Variable	Type	Description
name	string	Name of the migration resource (2-80 chars, alphanumeric + hyphens)
parent_id	string	Resource group ARM ID (/subscriptions/{sub}/resourceGroups/{rg})

Operation Control

Variable	Type	Default	Description
operation_mode	string	"discover"	Which operation to perform
source_machine_type	string	"VMware"	Source environment type (VMware / HyperV)
create_migrate_project	bool	false	Whether to create a new project
location	string	null	Azure region. Required only for create-project; otherwise read from the project

Project & Infrastructure

Variable	Type	Default	Description
project_name	string	null	Azure Migrate project name
connectivity_method	string	"Public-endpoint"	Public-endpoint or Private-endpoint
appliance_name	string	null	Appliance name for filtering discovery
source_appliance_name	string	null	Source appliance name (for fabric auto-discovery)
target_appliance_name	string	null	Target appliance name (for fabric auto-discovery)

cache_storage_account_id	string	null	Existing cache storage account ARM ID
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VM Replication

Variable	Type	Default	Description
machine_id	string	null	Source VM ARM ID from discovery
machine_name	string	null	Source machine internal name
os_disk_id	string	null	OS disk ID (simple mode)
disks_to_include	list(object)	[]	Disks (power user mode)
nics_to_include	list(object)	[]	NICs (power user mode)
target_vm_name	string	null	Name for the migrated VM

Target Configuration

Variable	Type	Default	Description
custom_location_id	string	null	Azure Local custom location ARM ID
target_hci_cluster_id	string	null	Target HCI cluster ARM ID
target_resource_group_id	string	null	Target resource group ARM ID
target_storage_path_id	string	null	Target storage container ARM ID
target_virtual_switch_id	string	null	Target logical network ARM ID (simple mode)

Migration & Removal

Variable	Type	Default	Description
protected_item_id	string	null	Protected item ARM ID (for get/migrate)
shutdown_source_vm	bool	false	Shut down source before migration
target_object_id	string	null	Protected item ARM ID (for remove)
force_remove	bool	false	Force removal of replication
job_name	string	null	Specific job name (jobs mode)
protected_item_name	string	null	Protected item name (get mode)

AVM Interface Variables

Variable	Type	Default	Description
enable_telemetry	bool	true	Toggle module telemetry
tags	map(string)	null	Resource tags
lock	object	null	Resource lock (kind, name)
managed_identities	object	{}	System/user-assigned

			identities
role_assignments	map(object)	{}	Role assignments to create
diagnostic_settings	map(object)	{}	Diagnostic settings to create

Output Reference

Discover Mode

Output	Description
discovered_servers	Filtered list: machine name, IPs, OS, boot type, OS disk ID
discovered_servers_count	Number of discovered servers with discovery data
total_machines_count	Total machines (including those without discovery data)
discovered_servers_raw	Full API response

Initialize Mode

Output	Description
replication_vault_id	Vault ARM ID
replication_vault_identity	Vault managed identity principal ID
replication_policy_id	Policy ARM ID
replication_extension_id / replication_extension_name	Extension ARM ID / name
cache_storage_account_id / cache_storage_account_name	Cache storage account
source_fabric_id / target_fabric_id	Resolved fabric ARM IDs
source_fabric_discovered / target_fabric_discovered	Discovered fabric details
replication_fabrics_available	All fabrics in the resource group

Replicate Mode

Output	Description
protected_item_id	Protected item ARM ID
protected_item_name	Protected item name
replication_state	Current replication health

Jobs Mode

Output	Description
replication_jobs	Summary of all jobs
replication_jobs_count	Total job count
replication_job	Details for a specific job
vault_id_for_jobs	Vault ID used for the query

List Mode

Output	Description
protected_items_summary	Summary of all protected items
protected_items_list	Complete raw list
protected_items_count	Total protected item count
protected_items_by_state	Items grouped by state
protected_items_by_health	Items grouped by health
protected_items_with_errors	Items with health errors

Get Mode

Output	Description
protected_item	Complete protected item details
protected_item_summary	Key protected item info
protected_item_custom_properties	Disk/network/fabric details
protected_item_health_errors	Health errors

Migrate Mode

Output	Description
migration_status	Operation status and VM names
migration_protected_item_details	Pre-migration item details
migration_validation_warnings	Pre-migration warnings
migration_operation_details	Raw async operation response

Remove Mode

Output	Description
removal_status	Operation status and confirmation
protected_item_details	Item details before removal
removal_operation_headers	Async operation headers

Common Outputs (all modes)

Output	Description
operation_mode	Current operation mode
migrate_project_id / resource_id	Azure Migrate project ARM ID
project_name_output	Project name
resource_group_id	Resource group ARM ID
location_output	Effective Azure region
machine_id	Machine ID being replicated
target_vm_name_output	Target VM name

Brownfield Support

The module works safely with existing Azure Migrate infrastructure, detecting and skipping creation of resources that already exist:

Resource	Detection Method
Replication Vault	Queries the DataReplication solution for an existing vaultId
Replication Policy	Lists existing policies and matches on instanceType
Replication Extension	Lists existing extensions and matches on expected naming pattern
Role Assignments	Lists assignments on cache storage account and checks for matching principal+role
Cache Storage Account	Uses cache_storage_account_id if provided, else reuses the project's recorded replicationStorageAccountId

You can safely run initialize multiple times — existing resources are reused, and only missing resources are created.

Support

- Terraform Registry: <https://registry.terraform.io/modules/Azure/avm-ptn-azure-local-migrate/azurerm/latest>
- AVM Documentation: <https://azure.github.io/Azure-Verified-Modules/>
- Azure Migrate Documentation: <https://learn.microsoft.com/azure/migrate/>

This guide is for the preview version of Azure Migrate for Terraform. Features and APIs may change in future releases.